

Ex# 19.2

vectors

Component form of vector

$$\overrightarrow{PQ} = v = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

Magnitude or Length of vector ($\overrightarrow{PQ}$)

$$|v| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example #01

Find the (a) Component form and (b) Length of the vector with initial point $P(-3, 4, 1)$ and terminal point $Q(-5, 2, 2)$

Sol:-

a) Component form

$$v_1 = x_2 - x_1 \Rightarrow -5 - (-3) = -2$$

$$v_2 = y_2 - y_1 \Rightarrow 2 - 4 = -2$$

$$v_3 = z_2 - z_1 \Rightarrow 2 - 1 = 1$$

$$\overrightarrow{PQ} = (-2, -2, 1)$$

coordinates
 $\begin{pmatrix} -5 \\ 2 \\ 2 \end{pmatrix}$
 $\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$

b) Length of $v = \overrightarrow{PQ}$

$$|v| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

$$= \sqrt{(-2)^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{4 + 4 + 1}$$

$$= \sqrt{9}$$

$$3$$

Example #02

A Small Cart is being pulled along a smooth horizontal floor with a 20-N force F making a 45° angle to the floor. what is the effective force moving the Cart forward?

Sol:-

$$\begin{aligned} a &= |F| \cos 45^\circ \\ &= 20 \times \cos 45^\circ \\ &= 20 \times 0.707 \\ &= 14.14 \text{ N} \end{aligned}$$

Example #03

Let $u = (-1, 3, 1)$ and $v = (4, 7, 0)$. find the Components

of

a) $2u + 3v$

b) $u - v$

c) $\left| \frac{1}{2} u \right|$

Sol:-

a) $2u + 3v$

$$2(-1, 3, 1) + 3(4, 7, 0) \Rightarrow (-2, 6, 2) + (12, 21, 0) \\ (10, 27, 2)$$

b) $u - v$

$$(-1, 3, 1) - (4, 7, 0) \Rightarrow (-5, -4, 1)$$

c) $\left| \frac{1}{2} u \right|$

$$\Rightarrow \sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2} = |v|$$

$$\Rightarrow \sqrt{(-1/2)^2 + (3/2)^2 + (1/2)^2}$$

$$\Rightarrow \sqrt{0.25 + 2.25 + 0.25}$$

$$\frac{1}{2} (-1, 3, 1) \Rightarrow \frac{-1}{2}, \frac{3}{2}, \frac{1}{2}$$

$$\sqrt{2.75}$$

$$1.658$$

Example # 04

Find a unit vector u in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$

$$v = v_1 i + v_2 j + v_3 k \Rightarrow \text{unit vector}$$

$$v_1 = m_2 - m_1$$

Sol:-

$$\vec{P_1 P_2} = v_1 i + v_2 j + v_3 k$$

$$= (3-1)i + (2-0)j + (0-1)k$$

$$v = 2i + 2j - 1k$$

$\frac{v}{|v|} \Rightarrow$ Direction vector.

$|v|$

Now find $|v|$

$$|v| = \sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2}$$

$$= \sqrt{(2)^2 + (2)^2 + (-1)^2} \Rightarrow \sqrt{4+4+1} = \sqrt{9} \Rightarrow 3$$

$$u = \frac{v}{|v|} = \frac{2i + 2j - k}{3} \Rightarrow \frac{2}{3}i + \frac{2}{3}j - \frac{1}{3}k$$

Example # 05

If $v = 3i - 4j$ is a velocity vector, express v as a product of its speed times its direction of motion.

Sol:-

Speed means length

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \\ &= \sqrt{(3)^2 + (-4)^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} = 5 \end{aligned}$$

unit vector for direction

$$\frac{v}{|v|} = \frac{3i - 4j}{5}$$

$$= \frac{3}{5}i - \frac{4}{5}j$$

Length and direction of vector

$$v = |v| \cdot \frac{v}{|v|}$$

$$\Rightarrow 5 \left(\frac{3}{5}i - \frac{4}{5}j \right)$$

$$\cancel{5} \times \frac{3}{\cancel{5}}i - \cancel{5} \times \frac{4}{\cancel{5}}j$$

$$\Rightarrow 3i - 4j$$

Ex# 12.2

In exercise 1-8, let $u = (3, -2)$ and $v = (-2, 5)$
find the a) component form and b) magnitude
(length) of the vector.

1. $3u$

Sol:-

$$3(3, -2)$$

$$9, -6$$

b) Length

$$\sqrt{(9)^2 + (-6)^2}$$

$$\sqrt{81 + 36} = \sqrt{117} = 3\sqrt{13}$$

02) $-2v$

Sol:-

$$-2(-2, 5)$$

$$= (4, -10)$$

b) Length

$$\sqrt{(4)^2 + (-10)^2}$$

$$\sqrt{16 + 100} = \sqrt{116} = 2\sqrt{29}$$

03) $u+v$

$$(3, -2) + (-2, 5)$$

$$= (1, 3)$$

b) Length

$$\sqrt{(1)^2 + (3)^2}$$

$$\sqrt{1 + 9} = \sqrt{10}$$

04) $u-v$

$$(3, -2) - (-2, 5)$$

$$(5, -7)$$

b) Length

$$\sqrt{(5)^2 + (-7)^2}$$

$$\sqrt{25 + 49} = \sqrt{74}$$

07) $\frac{3}{5}u + \frac{4}{5}v$

$$\frac{3}{5}(3, -2) + \frac{4}{5}(-2, 5)$$

$$\left(\frac{9}{5}, -\frac{6}{5}\right) + \left(\frac{-8}{5}, 4\right) = \left(\frac{1}{5}, \frac{14}{5}\right)$$

09- The vector $\vec{PQ}$ where $P = (1, 3)$ and $Q = (2, -1)$

Sol:-

$$v = \vec{PQ}$$

$$v = x_2 - x_1, y_2 - y_1$$

$$v = 2 - 1, -1 - 3$$

$$v = (1, -4)$$

10- The vector $\vec{OP}$ where O is the origin and P is the midpoint of segment RS , where $R = (2, -1)$ and $S = (-4, 3)$

Sol:-

$$\text{midpoint} \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{2 + (-4)}{2}, \frac{-1 + 3}{2} \right)$$

$$(-1, 1) \Rightarrow P \text{ Coordinates}$$

$$(0, 0) \Rightarrow O$$

Now find vector of $\vec{OP}$

$$v = \vec{OP}$$

$$v = x_2 - x_1, y_2 - y_1$$

$$v = 0 - (-1), 0 - 1$$

$$v = (1, -1)$$

11) The vector from the point $A = (2, 3)$ to the origin

Sol:-

$$O = (0, 0)$$

$$A = (2, 3)$$

$$V = x_2 - x_1, y_2 - y_1$$

$$V = 0 - 2, 0 - 3$$

$$V = (-2, -3)$$

12) The sum of $\vec{AB}$ and $\vec{CD}$ where $A(-1, -1)$ $B(2, 0)$
 $C(-1, 3)$ $D(-2, 2)$

Sol:-

$$V = \vec{AB}$$

$$V = x_2 - x_1, y_2 - y_1$$

$$V = 2 - (-1), 0 - (-1)$$

$$V = (1, 1)$$

$$V = \vec{CD}$$

$$V = x_2 - x_1, y_2 - y_1$$

$$V = (-2 - (-1), 2 - 3)$$

$$V = (-1, -1)$$

Now find sum

$$\vec{AB} = (1, 1)$$

$$\vec{CD} = (-1, -1)$$

$$1 + (-1)$$

$$1 + (-1)$$

$$0$$

$$0$$

$$(0, 0)$$

13- The unit vector that makes an angle $\theta = 2\pi/3$ with the positive x-axis.

unit vector with angle

$$u = (\cos \theta, \sin \theta)$$

Sol:-

$$\begin{aligned} u &= (\cos \theta, \sin \theta) \\ &= \left(\cos\left(\frac{2\pi}{3}\right), \sin\left(\frac{2\pi}{3}\right) \right) \end{aligned}$$

$$u = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$$

14- The unit vector that makes an angle $\theta = -3\pi/4$ with the positive x-axis

Sol:-

$$\begin{aligned} u &= (\cos \theta, \sin \theta) \\ &= \left(\cos\left(-\frac{3\pi}{4}\right), \sin\left(-\frac{3\pi}{4}\right) \right) \\ &= \left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2} \right) \end{aligned}$$

15) The unit vector obtained by rotating the vector $(0, 1)$ 120° counter clockwise about the origin.

Sol:-

Rotation of vector

$$\begin{aligned}(x', y') &= (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta) \\&= (0 \cos(120) - 1 \sin(120), 0 \sin(120) + 1 \cos(120)) \\&= 0 - \frac{\sqrt{3}}{2}, 0 + 1 - \frac{1}{2}\end{aligned}$$

$$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right)$$

16) The unit vector obtained by rotating the vector $(1, 0)$ 135° counterclockwise about the origin.

Sol:-

Rotation of vector

$$\begin{aligned}(x', y') &= (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta) \\&= (1 \cos(135) - 0 \sin(135), 1 \sin(135) + 0 \cos(135)) \\&= \frac{-\sqrt{2}}{2} - 0, \frac{\sqrt{2}}{2} + 0\end{aligned}$$

$$\left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$$

17) $\overrightarrow{P_1P_2}$ if P_1 is the point $(5, 7, -1)$ and P_2 is the point $(2, 9, -2)$.

Sol:-

$$\begin{aligned}\overrightarrow{P_1P_2} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (2 - 5, 9 - 7, -2 - (-1)) \\ &= (-3, 2, -1)\end{aligned}$$

$$V = V_1i + V_2j + V_3k$$

$$\begin{aligned}V &= -3i + 2j + (-1)k \\ &= -3i + 2j - 1k\end{aligned}$$

18) $\overrightarrow{P_1P_2}$ if P_1 is the point $(1, 2, 0)$ and P_2 is the point $(-3, 0, 5)$

Sol:-

$$\begin{aligned}\overrightarrow{P_1P_2} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (-3 - 1, 0 - 2, 5 - 0) \\ &= (-4, -2, 5)\end{aligned}$$

$$V = V_1i + V_2j + V_3k$$

$$V = -4i - 2j + 5k$$

19) $\overrightarrow{AB}$ if A is the point $(-7, -8, 1)$ and B is the point $(-10, 8, 1)$

Sol:-

$$\begin{aligned}\overrightarrow{AB} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (-10 - (-7), 8 - (-8), 1 - 1) \\ &= (-3, 16, 0) \\ &= -3i + 16j\end{aligned}$$

20) $\vec{AB}$ if A is the point $(1, 0, 3)$ and B is the point $(-1, 4, 5)$

Sol:-

$$\begin{aligned}\vec{AB} &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ &= (-1 - 1, 4 - 0, 5 - 3) \\ &= (-2, 4, 2)\end{aligned}$$

$$\begin{aligned}v &= v_1 i + v_2 j + v_3 k \\ &= -2i + 4j + 2k\end{aligned}$$

21) $5u - v$ if $u = (1, 1, -1)$ and $v = (2, 0, 3)$

Sol:-

$$\begin{aligned}5(1, 1, -1) - (2, 0, 3) \\ (5, 5, -5) - (2, 0, 3) \\ (3, 5, -8)\end{aligned}$$

$$\begin{aligned}v &= v_1 i + v_2 j + v_3 k \\ &= 3i + 5j - 8k\end{aligned}$$

22) $-2u + 3v$ if $u = (-1, 0, 2)$ and $v = (1, 1, 1)$

$$\begin{aligned}-2(-1, 0, 2) + 3(1, 1, 1) \\ (2, 0, -4) + (3, 3, 3) \\ (5, 3, -1)\end{aligned}$$

$$\begin{aligned}v &= v_1 i + v_2 j + v_3 k \\ v &= 5i + 3j - 1k\end{aligned}$$

Length and direction.

25) $2i + j - 2k$

Sol:-

$$V = 2i + j - 2k$$

$$\begin{aligned} |V| &= \sqrt{(V_1)^2 + (V_2)^2 + (V_3)^2} \\ &= \sqrt{(2)^2 + (1)^2 + (-2)^2} \\ &= \sqrt{4 + 1 + 4} \\ &= \sqrt{9} = 3 \end{aligned}$$

~~26) a~~

$$\frac{V}{|V|} = \frac{2i + j - 2k}{3} = \frac{2}{3}i + \frac{1}{3}j - \frac{2}{3}k$$

Length and direction

$$3 \left(\frac{2}{3}i + \frac{1}{3}j - \frac{2}{3}k \right)$$

26) $9i - 2j + 6k$

Sol:-

$$\begin{aligned} |V| &= \sqrt{(V_1)^2 + (V_2)^2 + (V_3)^2} \\ &= \sqrt{(9)^2 + (-2)^2 + (6)^2} \\ &= \sqrt{81 + 4 + 36} \\ &= \sqrt{121} \\ &= 11 \end{aligned}$$

$$\frac{V}{|V|} = \frac{9i - 2j + 6k}{11} = \frac{9}{11}i - \frac{2}{11}j + \frac{6}{11}k$$

Length and direction

$$|V| \times \frac{V}{|V|} = 11 \times \left(\frac{9}{11}i - \frac{2}{11}j + \frac{6}{11}k \right)$$

27) $5k$

Sol:-

$$\begin{aligned} |v| &= \sqrt{(v_1)^2} \\ &= \sqrt{(5)^2} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

$$\frac{v}{|v|} = \frac{5k}{5} = \frac{5}{5} k$$

Length and direction

$$|v| \times \frac{v}{|v|} \Rightarrow 5(k)$$

28) $\frac{3}{5}i + \frac{4}{5}k$

Sol:-

$$\begin{aligned} |v| &= \sqrt{(v_1)^2 + (v_2)^2} \\ &= \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} \\ &= \sqrt{\frac{9}{25} + \frac{16}{25}} \\ &= \sqrt{\frac{25}{25}} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

$$\frac{v}{|v|} = \frac{\frac{3}{5}i + \frac{4}{5}k}{1}$$

Length and direction

$$|v| \times \frac{v}{|v|} = 1 \left(\frac{3}{5}i + \frac{4}{5}k \right)$$

31) find the vectors whose lengths and directions are given. try to do the calculations without writing.

Sol:-

a) $2i$

b) $-\sqrt{3}k$

c) $\frac{1}{2} \left(\frac{3}{5}j + \frac{4}{5}k \right) \Rightarrow \frac{3}{10}j + \frac{4}{10}k$

d) $7 \left(\frac{6}{7}i - \frac{2}{7}j + \frac{3}{7}k \right) \Rightarrow 6i - 2j + 3k$

32)

32) find the vectors whose lengths and directions are given try to do the calculations without writing.

Sol:-

a) $-7j$

b) $\sqrt{2} \left(\frac{-3}{5}i - \frac{4}{5}k \right)$

c) $\frac{13}{12} \left(\frac{3}{13}i - \frac{4}{13}j - \frac{12}{13}k \right)$

d) $\frac{a}{\sqrt{2}}i + \frac{a}{\sqrt{3}}j - \frac{4a}{\sqrt{6}}k$